

# Service Operation



# Service Operation

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- The service operation stage is when the service is actually being delivered, and often is a much longer stage than the previous stages of strategy, design and transition



# Purpose of Service Operation

- The purpose of the service operation stage of the service lifecycle is to deliver the service at the level that was agreed through the service level management process
- This includes all the activities required to deliver the service as well as managing the technology used to deliver the service (such as applying updates, backing up data, and so on)
- Service operation must deliver the service effectively but also has to ensure that the cost of that delivery is within the operational costs that formed part of the original business case
- Should a service be operated at a higher cost than was originally envisaged, the benefits that were planned, such as cost savings, may never be realized



# Objectives of Service Operation

- It is the job of the service operation phase to deliver the service as agreed on in the SLA; this ensures that the business receives the level of service it expects
- Some service outages are inevitable; service operation will work to reduce both the number and impact of outages
- The service operation process of problem management aims to reduce repeat incidents that disrupt business activities, while incident management aims to resolve those incidents that do occur as quickly as possible
- Service operation is also responsible for controlling access to IT services. The access management process ensures that only authorized users can have access to the services provided



# Scope of Service Operation

- The scope of service operation described in the ITIL framework includes the “processes, functions, organization, and tools” that are used to deliver and support the agreed services
- This lifecycle stage is responsible for performing the critical day-to-day activities and processes that ensure the service meets the business requirement and enables the business to achieve its objectives
- It also collects the performance data that will be required by continual service improvement to identify and track improvement opportunities
- The ITIL Service Operation publication provides guidance on the successful management of the following:
  - **The Services Themselves**
    - All the activities required to deliver the services consistently within the agreed service levels
    - These activities may be carried out by the service provider, an external supplier, or the user or customer of that service
  - **The Service Management Processes**
    - The service operation processes of event, incident, problem and access management, and request fulfillment



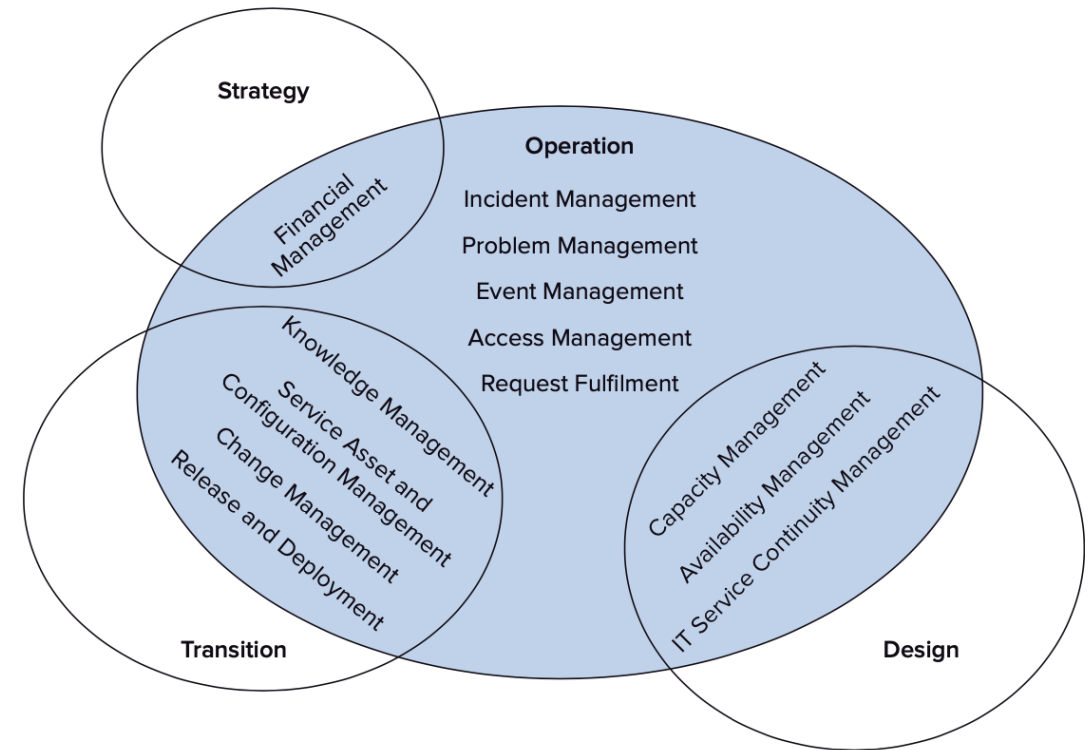
# Scope of Service Operation

## ■ The Technology Delivering

- IT services depends on the use of appropriate technology (network, desktops, servers, databases, data, monitoring tools, and so on)
- Service operation is responsible for managing the technology that delivers the services

## ■ The People

- Despite automation, service operation depends on the actions of the support staff members to ensure the service runs as it should
- Their management of the technology and processes is the key to successful service delivery



# Value it adds to the business

- Financial savings from reduced downtime as a result of the implementation of the service operation processes of problem and incident management
- Service operation includes the production of management information regarding the efficiency and effectiveness of the service delivery
- By carrying out the access management activities, service operation ensures that services are protected from unauthorized access, in accordance with the organization's security policy
- The request management process enables the business to obtain standard services quickly and with minimum bureaucracy
- By using technology to automate routine tasks, based on the information provided by the event management process, service operation reduces the number of staff members required to operate the service



# Processes





# Technical Management

- Whatever the name given to the team or teams in any particular organization (infrastructure support, technical support, network management, and so on), the function referred to in the ITIL framework as technical management is required to manage and develop the IT infrastructure
- This function covers the groups or teams that together have the technical expertise and knowledge to ensure that the infrastructure works effectively in support of the services required by the business
- It is responsible for managing the IT infrastructure. This would include ensuring that the staff members performing this function have the necessary technical knowledge to design, test, manage, and improve IT services
- Technical management staff members would be involved in drawing up the technical strategy and ensuring that the infrastructure can support the overall service strategy
- Technical management staff members would also carry out the technical design of new or changed services and would be involved in planning and implementing their transition to the operational environment



# Technical Management

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- Once the service is live, technical management provides technical support, resolving incidents, investigating problems, responding to alerts, and specifying any changes or updates required to have the service operate efficiently
- Technical management staff members will identify service improvements and work with the CSI manager to design, test, and implement these improvements



# Application Management

- This function shares many features with the technical management function, although in this case it is the application software that is supported and managed throughout its lifecycle, rather than the infrastructure
- Application management and applications development are not the same, and it is important to understand the differences between them:
  - Application management is involved in every stage of the service lifecycle, from ascertaining the requirements through design and transition and then operation and improvement
  - Application development is mostly involved in single, finite activities, such as designing and building a new service



# Service Desk Organization

- The best structure for the service desk is dependent upon the size and structure of the organization
- Global organization will have different needs from one with all its employees based in same location
- **Local Service Desk**
  - This option provides a service desk co-located with the users it serves; an organization with three offices would have three local service desks
  - There are advantages with this approach, in that the service desk is local, so it understands the local business priorities
- **Centralized Service Desk**
  - A more common structure for service desks is that of a centralized service desk
  - In this model, all users contact the same service desk
  - This has the benefit of providing economies of scale, because there is no duplication of provision
- **Virtual Service Desk**
  - The third organizational option described by ITIL is that of a virtual service desk
  - This option consists of two or more service desk locations that operate as one desk
  - Calls and emails are distributed across the staff members as if they were in one centralized location



# Incident Management

- In ITIL terminology, an incident is defined as an unplanned interruption to an IT service, a reduction in the quality of an IT service, or a failure of a CI that has not yet impacted an IT service (for example, failure of one disk from a mirror set)
- This is an important definition; because the incident is an interruption to service, restoring the service or improving the quality of the service to agreed levels resolves the incident
- Note that incident resolution does not necessarily include understanding why the fault occurred or preventing its recurrence; these are matters for problem management
- By understanding this distinction, you can see that resolving an incident does not need the skill that resolving a problem requires
- If the service can be restored by a simple reboot, then the user can be instructed to do this by the service desk staff, without involving the more skilled (and therefore more expensive) second-line technicians



# Managing Incidents

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- Incident Identification
- Incident Categorization
- Incident Prioritization
- Initial Diagnosis
- Incident Escalation
- Investigation and Diagnosis
- Resolution and Recovery
- Incident Closure



# Problem Management

- According to ITIL official terminology, a problem is defined as an underlying cause of one or more incidents
- Problem management is the process that investigates the cause of incidents and, wherever possible, implements a permanent solution to prevent recurrence
- Until such time as a permanent resolution is applied, it will also attempt to provide a workaround to enable the service to be restored and the incident to be resolved
- It is important to understand the differences between incidents and problems and to realize that an incident never becomes a problem
- Until and unless problem management is undertaken, incidents will recur, inconveniencing the business and occupying support staff time
- Problem management will reduce incidents, freeing up more time to undertake more problem management. It is a virtuous circle; the more time spent on it, the more time is freed up by it



# Problem Management Process

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- Detecting Problems
- Logging Problems
- Categorizing Problems
- Prioritizing Problems
- Investigating and Diagnosing Problems
- Identifying a Workaround
- Raising a Known Error record





# Event Management

- Modern infrastructure management depends to a large extent on the use of event monitoring tools
- These tools are able to monitor large numbers of configuration items simultaneously, identifying any issues as soon as they arise and notifying technical management staff
- The process of event management is responsible for managing events throughout their lifecycle
- Event management is one of the main activities of IT operations
- An event can be defined as any change of state that has significance for the management of a configuration item (CI) or IT service
- Many events are purely informational. Examples of informational events could include notification of a user logging onto an application (significant because the use of the application may be metered) or a transaction completing successfully (significant because the notification of the successful completion may trigger the start of the next transaction)



# Event Monitoring Tools

- **Active monitoring tools**

- Will poll devices to check that they are working correctly
- The tool will send a message and expect a positive response within a defined time, such as sending a “ping” to a device
- A failure to respond will be notified to support staff
- Some tools will have automated responses to such situations, perhaps automatically restarting a device or rerouting data to avoid the faulty CI so that the service is not affected

- **Passive monitoring tools**

- Do not send out polling messages
- They detect events generated by CIs and correlate them (that is, identify related events)



# Event Management vs Event Monitoring

- It is important to understand the difference between the two similar but different activities of monitoring and managing events
  - **Event management**
    - It is concerned with generating events and detecting notifications that have been produced
    - These events are produced so that they can be monitored
    - They provide useful information regarding the status of the infrastructure
  - **Monitoring**
    - Detects these notifications but goes further than this
    - Monitoring includes actively checking CIs to ensure that they are working as they should, whether or not an event has been generated



# Access Management

- Access management is the process of granting authorized users the right to use a service while preventing access to nonauthorized users
- It also is sometimes referred to as rights management or identity management
- Access requirements can change frequently, and service operation is responsible for granting access quickly, in line with the needs of the business, while ensuring that all requests are properly authorized
- Manage access to services, carrying out the policies defined within information security management
- Ensure that all requests for access are verified and authorized. This may include requests to restrict or remove access.
- Ensure that requests are dealt with efficiently, balancing the requirement for authorization and control with the need to be responsive to business requirements
- Ensure that the rights that have been granted are used in accordance with security policies

